

4.4 Indeterminate Forms and L'Hospital's rule (Page 304 of the text book)

You may remember that some limits can be challenging to calculate, especially the limit of the quotient of two functions. In most cases, one has to do something special to find the limit, like in the following example:

Example 1 $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9} \left(\frac{0}{0} \right)$

As you can see, both the numerator and the denominator tend to 0. We call this an Indeterminate Form of Type $\frac{0}{0}$ and indicate that next to the expression.

In this case we can find the limit by using factorisation, as follows:

$$\begin{aligned} & \lim_{x \rightarrow 3} \frac{x-3}{x^2-9} \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 3} \frac{x-3}{(x-3)(x+3)} \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 3} \frac{1}{x+3} \quad (\text{we may divide the numerator and denominator by } x-3 \text{ since } x-3 \neq 0 \text{ when } x \rightarrow 3) \\ &= \frac{1}{6} \end{aligned}$$

Another example is $\lim_{x \rightarrow 0} \frac{\sin x}{x}$:

We used our knowledge of the trigonometric functions to show that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

Example 2 $\lim_{x \rightarrow \infty} \frac{x^5+5x^3}{x^4+x^2} \left(\frac{\infty}{\infty} \right)$

As you can see, both the numerator and the denominator tend to ∞ . We call this an Indeterminate Form of Type $\frac{\infty}{\infty}$ and indicate that next to the expression.

In this case we can again find the limit by using factorisation, as follows:

$$\begin{aligned} & \lim_{x \rightarrow \infty} \frac{x^5+5x^3}{x^4+x^2} \left(\frac{\infty}{\infty} \right) \\ &= \lim_{x \rightarrow \infty} \frac{x^4(x+5/x)}{x^4(1+1/x^2)} \left(\frac{\infty}{\infty} \right) \\ &= \lim_{x \rightarrow \infty} \frac{(x+5/x)}{(1+1/x^2)} \quad (\text{We may divide the numerator and denominator by } x^4 \text{ since } x^4 \neq 0 \text{ when } x \rightarrow \infty) \\ &= \infty \quad (\text{The denominator tends to } 1 \text{ while the numerator tends to } \infty.) \end{aligned}$$

But there are limits that we cannot find with our current knowledge, such as $\lim_{x \rightarrow 1} \frac{\ln x}{x-1} \left(\frac{0}{0} \right)$.

In cases like these, there is a very useful tool, namely L'Hospital's rule:

L'Hospital's rule

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, provided that the limit on the right exists (or is ∞ or $-\infty$)

Note:

1. L'Hospital's rule is also valid for one sided limits and limits at infinity.
2. The rule states that the limit of the quotient of the functions is equal to the limit of the quotient of the derivatives, which in general is NOT equal to the derivative of the quotient!

Here are some examples:

Example 3 $\lim_{x \rightarrow 1} \frac{\ln x}{x-1} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 1} \frac{1/x}{1} \text{ (L'Hospital's rule)}$

$$= \lim_{x \rightarrow 1} \frac{1}{x} = 1$$

Example 4 $\lim_{x \rightarrow \infty} \frac{\ln x}{x^2} \left(\frac{\infty}{\infty} \right) = \lim_{x \rightarrow \infty} \frac{1/x}{2x} \text{ (L'Hospital's rule)}$

$$= \lim_{x \rightarrow \infty} \frac{1}{2x^2} = 0$$

Be careful! You cannot use L'Hospital's rule for every limit of a quotient! Here are two examples:

Example 5 $\lim_{x \rightarrow -\infty} \frac{\ln x}{x^2}$ does not exist, because $\ln x$ is only defined for $x > 0$.

Example 6 $\lim_{x \rightarrow -\infty} \frac{3^x}{x^2} = 0$ Note that this is not an indeterminate form: The numerator tends to 0 and the denominator tends to ∞ .

Example 7 $\lim_{x \rightarrow 0} \frac{2x}{e^{3x}-1} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{2}{3e^{3x}} \text{ (L'Hospital's rule)}$

$$= \frac{2}{3}$$

Example 8 $\lim_{x \rightarrow \pi/2} \frac{\cos x}{2x-\pi} \left(\frac{0}{0} \right) = \lim_{x \rightarrow \pi/2} \frac{-\sin x}{2} \text{ (L'Hospital's rule)}$

$$= -\frac{1}{2}$$

There are other limits where we can also apply L'Hospital's rule, for example indeterminate products:

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ or $-\infty$, then it is not clear what the limit of the product will be. We call this an indeterminate form of type $(0 \cdot \infty)$

In cases like these, we write the product as a quotient, $f(x)g(x) = \frac{f(x)}{1/g(x)}$ or $f(x)g(x) = \frac{g(x)}{1/f(x)}$

Here is an example:

Example 9 $\lim_{x \rightarrow \infty} x e^{-x} (\infty \cdot 0) = \lim_{x \rightarrow \infty} \frac{x}{e^x} \left(\frac{\infty}{\infty}\right)$

$$= \lim_{x \rightarrow \infty} \frac{1}{e^x} \quad (\text{L'Hospital's rule})$$

$$= 0$$

Example 10 $\lim_{x \rightarrow 0^+} (\sin x)(\ln x)$ HOMEWORK!

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = -\infty$, then it is not clear what the limit of the sum will be. We call this an indeterminate form of type $(\infty - \infty)$.

In cases like these, we try to convert the difference into a quotient, or we factorise or use algebraic manipulation to see if we can find the limit. Here are some examples:

Example 11 $\lim_{x \rightarrow \infty} [\ln(x) - \ln(x+1)] (\infty - \infty) = \lim_{x \rightarrow \infty} \ln\left(\frac{x}{x+1}\right)$

$$= \lim_{x \rightarrow \infty} \ln\left(\frac{1}{1+1/x}\right) = \ln(1) = 0$$

Example 12 $\lim_{x \rightarrow 0^+} \left[\frac{1}{x} - \frac{1}{x+1}\right] (\infty - \infty) = \lim_{x \rightarrow 0^+} \left(\frac{(x+1)-x}{x(x+1)}\right)$

$$= \lim_{x \rightarrow 0^+} \left(\frac{1}{x(x+1)}\right) = \infty$$

Example 13 $\lim_{x \rightarrow -\infty} [x^4 + x^3] (\infty - \infty) = \lim_{x \rightarrow -\infty} x^3(x+1)$

$$= \infty$$

Indeterminate powers: Several indeterminate forms arise from the limit $\lim_{x \rightarrow a} f(x)^{g(x)}$. The types that arise are denoted by 0^0 , ∞^0 or 1^∞ .

In these cases we follow a similar strategy to that of logarithmic differentiation: Let's look at an example:

Example 14 $\lim_{x \rightarrow \infty} (x)^{1/x} (\infty^0)$

Let $y = (x)^{1/x}$. Then $\ln y = \ln(x)^{1/x}$ and so

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{1}{x} \ln[x] = \lim_{x \rightarrow \infty} \frac{\ln x}{x} \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{1/x}{1} \quad (\text{L'Hospital's rule})$$

$$= \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

It now follows that $\lim_{x \rightarrow \infty} (x)^{1/x} = e^0 = 1$

Example 15: $\lim_{x \rightarrow \infty} (1 + 4/x)^x (1^\infty)$ HOMEWORK!